

Multivariable Calculus

(MT2008)

Date: 31 December 2024

Course Instructor

Muhammad Yaseen

Final Exam

Total Time (Hrs): 3

Total Marks: 100

Total Questions: 3

Section

Student Signature

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Attempt all the questions.

CLO #1. Formulate the equation of lines, planes and surfaces.

Q1: Write standard definition of Quadric Surfaces. Identify and sketch the given surface

$4x^2 - y^2 + 2z^2 + 4 = 0$ [10 marks].

CLO #2. Calculate the rate of change of multivariable functions and its applications.

Q2: (a). Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ if $x^3 + y^3 + z^3 + 6xyz = 1$.

$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z}$ $\frac{\partial z}{\partial y} = -\frac{F_y}{F_z}$

(b). Find the local maximum and minimum values and saddle points of the given function

$f(x, y) = x^4 + y^4 - 4xy + 1$.

(c). Find the tangent plane to the elliptic paraboloid $z = 2x^2 + y^2$ at the point $(1, 1, 3)$.

$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$

[5+10+5 marks].

CLO #3. Evaluate the integral of a multivariable function and its applications.

Q3: (a). Evaluate the iterated integral $\int_0^1 \int_y^1 \sin x^2 dx dy$.

(b). Find the flux of the vector field $F(x, y, z) = xk + yj + zi$ across the unit sphere $x^2 + y^2 + z^2 = 1$.

(c). Use the Divergence theorem to calculate the surface integral $\iint_S F \cdot dS$; that is, calculate the flux of F across S , $F(x, y, z) = xye^z i + xy^2z^3 j - ye^z k$, S is the surface of the box bounded by the coordinate planes and the planes $x = 3$, $y = 2$ and $z = 1$.

(d). Define curvature and also find the curvature of a circle of radius a .

(e). Find a parametrization for the surface S formed by the part of the hyperbolic paraboloid $z = y^2 - x^2$, lying inside the cylinder of radius one around the z -axis and for the boundary curve C of S . Then verify Stokes Theorem for S using the normal having positive k -component and the vector field $F = y \mathbf{i} - x \mathbf{j} + x^2 \mathbf{k}$. Stokes -

(f). Find the volume of the solid that lies under the paraboloid $z = x^2 + y^2$, above the xy -plane, and inside the cylinder $x^2 + y^2 = 2x$. $\iiint \frac{1}{2} dz dx dy$

(g). Integrate $G(x, y, z) = \sqrt{1 - x^2 - y^2}$ over the "football" surface S formed by rotating the curve $x = \cos z, y = 0, -\pi/2 \leq z \leq \pi/2$, around the z -axis.

(h). Evaluate the surface integral, $\iint (x + y + z) dS$, where S is the part of the half cylinder $x^2 + z^2 = 1, z \leq 0$, that lies between the planes $y = -2$ and $y = 0$. $\frac{1}{2}$

$\int_0^1 \int_{-2}^0 \frac{1}{2} dy dz$ [5+10+10+5+10+10+10+10 marks]

$\frac{1}{2} \int_0^1 \int_0^1 dz dy$

$z=0$